

Arunabha Majumder

Automation & Robotics Engineer | Proof-of-Concept | Prototyping & Testing | Sensors & Control

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Portfolio: arunabha22.github.io/arunabha-majumder | Project: www.viexo.aau.dk

SUMMARY

Automation and robotics engineer (MTech Mechatronics, BTech Mechanical Engineering, PhD in progress) who takes ideas from concept and feasibility to working prototypes and tested hardware. Hands-on with robotic mechanisms, actuators, sensors, pneumatics and control systems. Clear communicator across technical and non-technical audiences in international teams in Denmark and India.

SKILLS

- **Automation & Robotics:** Robotics, actuators, pneumatics, sensors, motors, encoders, DAQ, embedded control (Arduino), LabVIEW
- **Innovation:** Concept creation, feasibility studies, proof-of-concept, prototyping, testing, validation
- **Control & Software:** PID and model-based control, kinematics, Jacobian-based force estimation, MATLAB/Simulink, Python
- **Design & Build:** SolidWorks (CSWA), Autodesk Fusion 360, FEA, 3D printing, machining, assembly, custom test rigs
- **Communication:** Stakeholder collaboration, technical communication, teaching, English (C1)

EXPERIENCE

Research Engineer – Robotics & Automation (PhD Project: VIE XO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Took a hybrid-actuated shoulder exoskeleton from concept to tested prototype; cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Evaluated a sensing approach on a 5-bar planar parallel robot: end-effector force estimated from variable-stiffness actuators via the Jacobian, validated against a force sensor to below 10% RMS error.
- Integrated motor, encoder, force sensors, DAQ and embedded control; built custom automated test rigs.
- Designed and tested an Assist-as-Needed controller, reducing user muscle effort by more than 15%.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

MTech Researcher – Pneumatic Actuator Systems | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Built pneumatic artificial muscles and a pneumatic test set-up (valves, pressure sensors, DAQ).
- Developed a neural-network gain-scheduled PID controller: tracking error 0.3–0.78%.

SELECTED PROJECTS

- **Rehabilitation Exoskeleton (2018–2019):** Concept, CAD, fabrication and user testing; Best Major Project Award and INR 100,000 TATA Technologies Innovation Grant.
- **Efficycle (2018):** Multidisciplinary team build of a human-electric hybrid vehicle.
- **Go-Kart (2017):** Designed, fabricated and assembled from the ground up.

EDUCATION

- **PhD Candidate, Mechatronics and Robotics** | Aalborg University, Denmark | 2023 – Present
- **MTech, Mechatronics** | AcSIR – CSIR-CMERI, India | 2020 – 2022
- **BTech, Mechanical Engineering** | Siddaganga Institute of Technology (VTU), India | 2015 – 2019

PUBLICATIONS & AWARDS

- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer. Best Research Paper Award.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.
- Certified SolidWorks Associate (CSWA)

LANGUAGES & REFERENCES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

References available upon request.